Department of Computer Science
PhD project: Developing Artificial Intelligence Approaches to Enhance Representations of Turbulence in Atmospheric Models

Neural subgrid closures coupled to MONC: stability, fidelity and cost of emulated turbulence at kilometre scale

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AFESP Annual Conference 2026 · PhD progress **From offline emulators to closures that drive MONC: stable for month long simulations, close in temperature reproduction, different in cloud fields, and now learning/fine-tuning online.**

Research roadmap

● completed ◑ ongoing ○ planned

Data generation and selection	Network design	Physics-guided vs data-driven	Multi-task framework	Offline training and evaluation	Online coupling	Analysis and generalisation	Online training and fine-tuning	Future directions
● RCE and ARM LES, regime-balanced → 2	● stencil inputs matched to Smagorinsky → 2	● constrained vs data-driven CNNs (EGU 2025)	● Ri and regime heads, 18 designs → 3	● full-domain, cross-regime tests → 3	● FTorch, SmartSim, ENNUF → 7	● ensembles, same-state test → 5 6	● streamed fine-tuning → 8	○ multi-regime pre-training
◑ new regimes: Sc, neutral and stable BL, fog, CAO	● MLP, ResMLP, TabTransformer → 3	○ neural operators; ablations	● uncertainty task weighting → 3	◑ data scaling	● stable coupled runs → 4	◑ 30-day runs → 4	◑ active learning	○ curriculum; hybrid models

Numbers point to the panels below.

1 Why emulate the turbulence closure?

- At kilometre scale the subgrid-scale (SGS) closure carries much of the turbulent transport and shapes entrainment, cloud depth and cover (Honnert et al. 2020).
- Traditional SGS schemes may be inaccurate and too simplistic.
- Neural emulators have shown promise, yet many destabilise once coupled; online evaluation is the decisive test (Ross et al. 2023).
- **We emulate the coefficients Km and Kh, so MONC keeps its own down-gradient diffusion operator.**

$$K_{m,h} = \lambda^2 S\, f_{m,h}(\mathrm{Ri}), \quad \mathrm{Ri} = (g/\theta_0)\, \partial\theta/\partial z \;/\; |\partial u/\partial z|^2$$

Ri ≤ 0	0 < Ri < 0.25	Ri ≥ 0.25
Unstable: more mixing	Transitional: steep cut	Supercritical: no mixing

The three Ri regimes are also the emulators' classification task and the sampling strata.

2 Sparse, stratified training data

RCE · deep convection
1 km, 64 × 64 × 99, 27 snapshots
24.4% used
2.67 M of 10.95 M

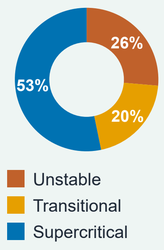
ARM · shallow cumulus
50 m, 192 × 192 × 220, 9 snapshots
1.9% used
1.38 M of 73 M

4.05 M samples · 4.8% of 83.9 M
66% RCE, 34% ARM · ~60 k per regime per snapshot

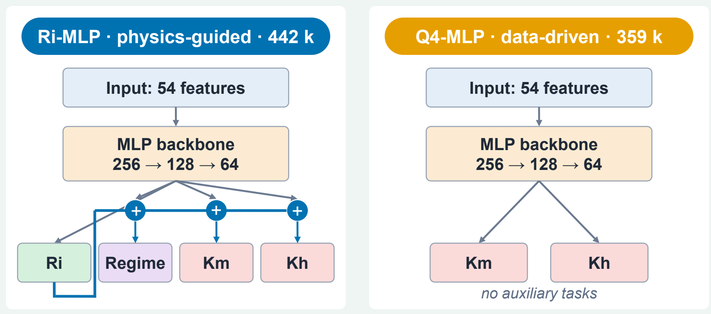
54 input features per point



Ri-regime balance

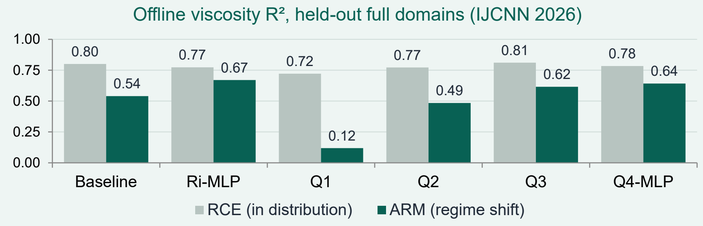


3 Physics-guided multi-task emulators



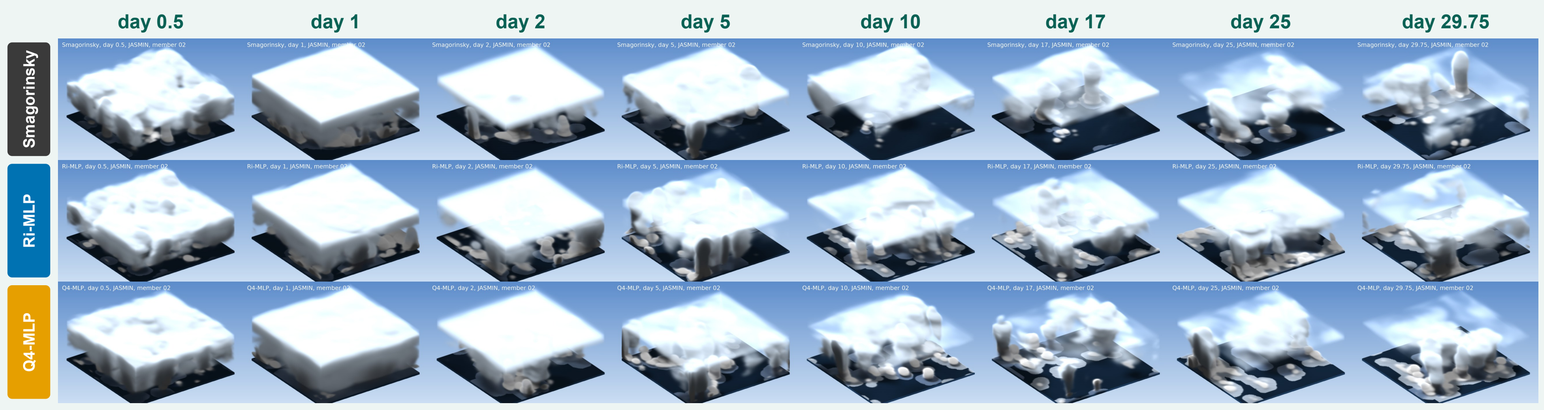
Predicted Ri conditions the regime and coefficient heads. Loss: $\mathbf{L = \Sigma_k\, [L_k/2\sigma_k^2 + \log \sigma_k]}$, learned task uncertainties σk (Kendall et al. 2018).

Offline viscosity R², held-out full domains (IJCNN 2026)

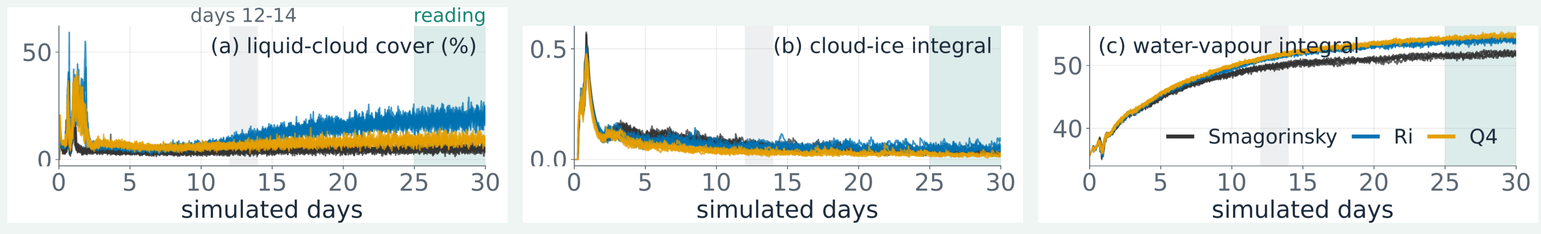


- Uncertainty weighting won all 54 configurations (+23–33% R² over manual).
- Ri conditioning transfers best to ARM; active-turbulence viscosity R² 0.43 vs 0.17 (+158%).
- Residual and attention backbones inflate variance up to 5× under the shift.

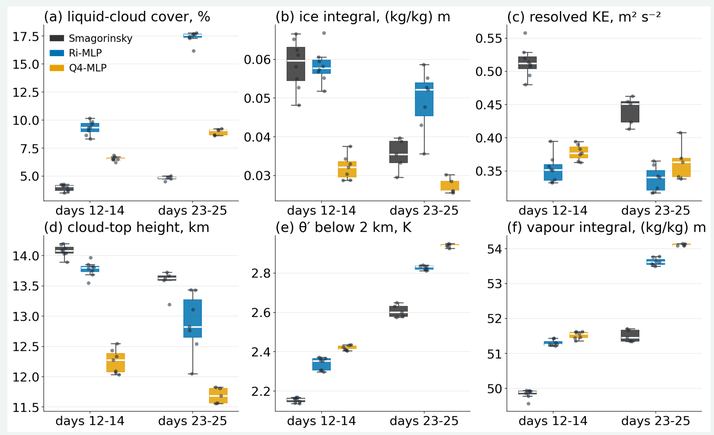
4 Coupled runs are stable for a month: day 0.5 to day 30 in 3-D



24 of 24 runs stable with no safeguards (8 initial-condition members × 3 closures); 15 of 24 have reached day 30; also stable at 0.5–2 km and at +2 and +4 K.
Top: member 02 in 3-D (colour marks phase, not amount; qualitative). Below: every member over 30 days; shaded, days 12–14 and the days 25–30 reading window, read after spin-up under a rule fixed in advance. RCE, 1 km, 64 × 64 km, CASIM, JASMIN.

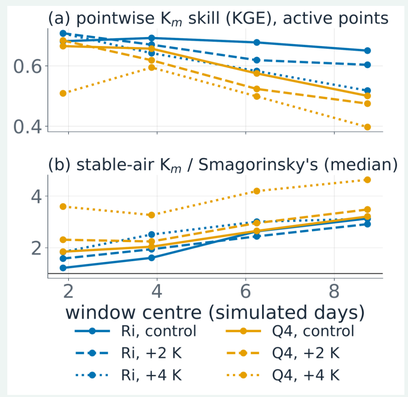


5 Temperature close, clouds depart



Liquid cloud **+134% / +65%** (Ri / Q4) at days 12–14, **+261% / +86%** at days 23–25, every member outside the Smagorinsky range; θ′ +0.2–0.3 K and vapour +3–5% stay close. Members: 8/8/7, then 5/7/5; both windows precede equilibrium.

6 Same states: physics guidance holds



Smagorinsky drives the model for 10 days; each emulator computes K on the identical state.

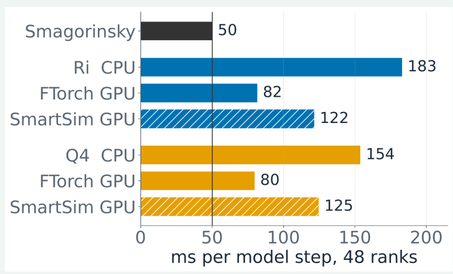
Km KGE	Ri	Q4
control	**0.69**	0.61
+2 K	**0.67**	0.58
+4 K	**0.63**	0.54

Ri-MLP is closer in all 12 climate × time windows, most at +4 K; both drift as the state leaves the training hours.

7 Coupling: three modes, three backends

MONC state u, v, w, θ, q per column → **54 features** stencil in Fortran → **Emulator** 3 backends, CPU or GPU → **Km, Kh** native arrays → **MONC diffusion** unchanged

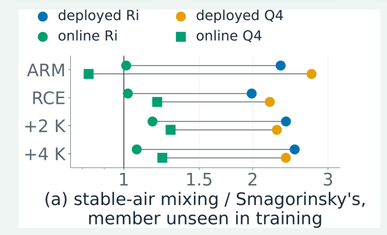
Smagorinsky native, to round-off · **Emulator-driven** emulator closes the model · **Dual-run** scored on identical state

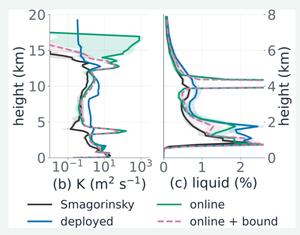


FTorch fastest, in process
SmartSim decouples model and inference; streams data
ENNUF pure Fortran
Throughput vs Smagorinsky: 0.28–0.35× CPU, 0.45–0.64× GPU (RACC2)

8 Learning online while MONC runs

MONC on CPUs RCE, +2 K, +4 K, ARM → **SmartRedis** live inputs and labels → **Replay buffer** regime, K, height strata → **GPU trainer** one A100, T18 recipe





T18 recipe: warm start from deployed weights; hurdle head (K = 0 where no mixing); 50% rehearsal; loss on K and log K; stratified replay. Labels always come from a Smagorinsky-driven run; nothing is written to disk.

Stable-air over-mixing 2.0–2.8× → 0.8–1.3×. Coupled, the hurdle plus a mixing-length bound ($K \le l^2|S|$) ends the month-long cooling at the lid; at +4 K the lowest stratosphere still cools.

Conclusions

1. **Online coupled simulations are stable** — tested for 30 days, on unseen grids and in warmer climates
2. **Temperature right, cloud fields vary** — 2–3.5× Smagorinsky's low liquid cloud
3. **Physics guidance helps** — closer to the reference on identical states, most under warming
4. **Multiple backends on hybrid compute** — FTorch, SmartSim and ENNUF on CPU and GPU
5. **Long runs reveal hidden problems** — and a fix: hurdle head plus a mixing-length bound

Where this sits in AFESP Theme 2

2A, 2B km-scale ML parametrisation — what the emulators reproduce, and where they extrapolate (1–6)

2C computing — one component, three backends, measured cost; streaming for online learning (7, 8)

2D ensembles — initial-condition ensembles as the noise floor; a 30-day reading window (4, 5)

References
1. Panda et al., Physics-guided multi-task learning for subgrid scale turbulence parameterization, IJCNN 2026, IEEE. doi:10.1109/IJCNN69854.2026.11728747
2. Panda et al., Physics-aware multi-task learning for atmospheric turbulence parameterization, PASC '26, ACM. doi:10.1145/3815572.3815750

Acknowledgements: AFESP, University of Reading; Met Office (CASE studentship, HPC); JASMIN and ORCHID; RACC2; ICCS, Cambridge (FTorch).

Presented at
- IEEE WCCI / IJCNN 2026, Maastricht · oral
- PASC 2026, Bern · talk
- km-Scale Global Modelling Summit 2026, Hamburg · poster
- ICCS Summer School 2026, Cambridge · poster
- AFESP Conference 2025, Reading · talk
- EXCLAIM 2025, ETH Zurich · poster
- EGU General Assembly 2025 · oral

ICCS Summer School 2026
Best Poster, runner-up

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IJCNN DOI · IJCNN data · IJCNN code · PASC DOI · PASC data · PASC code